Deterministic Admissibility Engineering: Mechanized Type Theory, Bare-Metal Clamps, and the Regeneration Gap

Formal Microarchitectural Translation Bridge

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Target Baseline: Phase 7 Realization (v1.3.1-exec) — Bounds: $\tau_{\text{override}} \leq 11.99\,\text{ms}$, $C_{\text{ops}} = 0$
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1 Executive Summary & Foundational Axioms

Autonomous agent frameworks driving non-deterministic generative models systematically undermine classical distributed systems idempotency. While conventional network architectures assume that retry loops replay identical request payloads, autoregressive generative agents re-synthesize arguments upon encountering network timeouts or ambiguous states. This semantic drift creates a critical distributed failure: **The Regeneration Gap**, wherein syntactically divergent requests representing identical underlying business intents bypass transport-layer deduplication keys, inducing catastrophic duplicate mutations.

This specification formalizes **Deterministic Admissibility Engineering (DAE)**—a mathematical, microarchitectural, and operational framework built on a single governing axiom:

AXIOM OF ADMISSIBILITY: Existence $\not\equiv$ Permission

Under DAE, no instruction, neural network inference, or external API execution may commit until its state vector passes a pre-execution admissibility gate verified at the kernel and hardware boundary.

Dual-Lens Model Paradigm Comparison

```
PERMISSIVE MODEL (Legacy Agent & Cloud Architecture):
[State Change Trigger] ---> [Autoregressive Synthesis] ---> [Execute API / Tool] (High Risk)
                                    |                              |
                               (Timeout)                   (External Mutation)
                                    |                              |
                          [Re-Synthesis] ---> (Hash Mismatch) ---> [Duplicate Commit]

DETERMINISTIC ADMISSIBILITY MODEL (DAE Architecture):
[State Change Trigger] ---> [Pre-Ingress Gate (eBPF/XDP)] ---> [PASS: Commit <= 1] (Zero Entropy)
                                    |
                                    v
                  [Short-Circuit: Poset L (POSIX 40 / Sub-12ms)]
```

2 Formal Logic and Automated Verification: Coq, Lean, and Z3

To evaluate how DAE achieves mathematical certainty without reliance on heuristic solvers or ambiguous narrative specifications, the underlying formal mechanisms must be contrasted with alternative theorem-proving environments.

Table 1: Comparative Taxonomy of Automated Verification Paradigms and the DAE Runtime Substrate

Dimension	**Z3 SMT Solver**	**Lean 4 Environment**	**Coq / Rocq (CIC)**
Logical Core	First-Order Logic + Theories (DPLL/CDCL)	Dependent Type Theory (Calculus of Constructions)	Calculus of Inductive Constructions (CIC Micro-Kernel)
Solving Method	Push-button automated satisfiability search	Interactive proof construction / Functional language	Interactive tactic proof elaboration checking explicit proof terms
Axiomatic Footprint	Heuristic decision procedures, theory solvers	Dependent kernel; Mathlib frequently assumes classical axioms	Empty-Axiom Verification: `Print Assumptions` $\equiv \emptyset$
Hardware Bridge	Static symbolic bounds check in software	Verified code generation via C backend	Unbroken vertical functor $\Phi : \mathcal{L} \to \mathcal{E}$ to bare-metal clamps
Architectural Role	Automated counterexample search	Pure mathematical formalization (Mathlib)	Active runtime root: proof-closed kernel enforcing state clamps

While automated SMT solvers like Z3 verify bounded software properties efficiently, they rely on complex decision procedures that cannot easily express higher-order categorical mappings. Lean 4 excels in pure mathematics, but its ecosystem frequently imports classical axioms. Coq was selected for DAE because its small, 30-year hardened micro-kernel allows strict enforcement of the zero-axiom criterion: every theorem in `dae_invariants.v` compiles to explicit lambda terms with zero unproven postulates (Assumptions(T) $\equiv \emptyset$) and zero skipped proofs (`Admitted.` $\equiv 0$).

3 The Regeneration Gap & Invariant Bounding

3.1 Formal Definition of the Regeneration Gap

Let $G_\theta(c, \xi)$ be an autoregressive generative model parameterized by weights θ, operating over conversational context c, under stochastic perturbation $\xi \sim \mathcal{D}$. Let Ω represent the physical or economic state space, and let $f : \mathcal{A} \times \Omega \to \Omega$ represent an effectful action invocation.

Definition 1 (**The Regeneration Gap**)**.** Let I be a discrete business intention. A Regeneration Gap occurs when an agent attempts to execute I over an unreliable channel, encounters an ambiguous

completion outcome $U \in \{\text{Timeout}, \text{Dropped Ack}\}$, and invokes G_θ to generate a remedial attempt, producing distinct payloads $a_1, a_2 \in \mathcal{A}$ such that:

$$a_1 \sim_{\text{intent}} a_2 \quad \wedge \quad a_1 \neq a_2 \implies \text{Key}(a_1) \neq \text{Key}(a_2) \tag{1}$$

resulting in duplicate external side-effects across execution attempts $E(I)$:

$$\sum_{e \in E(I)} \mathbf{1}[\text{Committed}(e)] > 1 \tag{2}$$

3.2 The Single-Commit Invariant

To eliminate the Regeneration Gap, execution authority is moved entirely outside the generative model and bound to an immutable execution membrane:

Invariant 1 (**Single-Commit Invariant**)**.** For every authorized logical operation $o \in \mathcal{O}$ registered within the operational substrate, the sum of all committed side-effects across all retries, regenerations, or duplicate frames through all time t is strictly bounded by unity:

$$\forall o \in \mathcal{O}, \quad \forall t \geq 0, \quad \sum_{e \in E(o)} \mathbf{1}[\text{Committed}(e, t)] \leq 1 \tag{3}$$

4 The 4-Tier Vertical Architectural Hierarchy

DAE realizes the Single-Commit Invariant through four vertically integrated, mechanically verified layers spanning from pure type theory to physical actuators.

4.1 Tier 1: Empty-Axiom Interactive Proof Theory ($\mathcal{L}$)

The foundation is formalized in the Coq Interactive Theorem Prover (Calculus of Inductive Constructions). To prevent specification hand-waving, proofs must maintain an **empty-axiom discipline**:

$$\text{Assumptions}(\texttt{dae_invariants.v}) \equiv \emptyset \tag{4}$$

The carrier set $\mathcal{L}$ is defined as a bounded join-semilattice:

$$\mathcal{L} = \{\bot, \ell_1, \ell_2, \ell_3, \top\} \quad \equiv \quad \{\text{PASS}, \text{PWC}, \text{CORRUPT}, \text{REFUSAL}, \text{FAIL}\} \tag{5}$$

where $a \sqcup_{\mathcal{L}} b = \sup(a, b)$.

Theorem 1 (**Absorbing Supremum Closure**)**.** For any set of concurrent evaluations $S \subseteq \mathcal{L}$, the evaluation operator $\mathcal{L}_{\text{eval}}(S)$ resolves deterministically to the supremum (least upper bound) of violations:

$$\mathcal{L}_{\text{eval}}(S) = \sup_i \{s_i \in S\} \tag{6}$$

If a single invariant rule emits $\top$, the entire batch resolves to $\top$ under constructive induction with zero axiomatic admissions.

4.2 Tier 2: The Order-Preserving Functor ($\Phi : \mathcal{L} \to \mathcal{E}$)

Algebraic lattice states are mapped to standardized POSIX execution signals via an order-preserving covariant functor Φ:

$$\Phi : \mathcal{L} \to \mathcal{E} \tag{7}$$

Per normative erratum `ARC-ERR-2026-001`, code 10 (`FREEZE`) is reserved as an internal scheduling yield outside $\mathrm{Im}(\Phi)$. The operational dominance chain enforces linear priority:

$$\Phi(\top) = 40 \succ \Phi(\ell_3) = 32 \succ \Phi(\ell_2) = 30 \succ \Phi(\ell_1) = 12 \succ 10 \succ \Phi(\bot) = 0 \tag{8}$$

Table 2: Lattice State & Microarchitectural Action Table

Algebraic Element	**POSIX Vector**	**Operational Behavior**	**Microarchitectural Action**
$\top$ (FAIL)	POSIX 40	Complete Execution Halt	Trigger Hardware NMI
ℓ_3 (REFUSAL)	POSIX 32	Wire-Speed Ingress Drop	eBPF XDP DROP Ring 0 Filter
ℓ_2 (CORRUPT)	POSIX 30	Ledger State Invalidation	Atomic Memory Barrier Reset
ℓ_1 (PWC)	POSIX 12	Pending Worker Clearance	Hold Pending Out-of-Band Token
– (FREEZE)	POSIX 10	Runtime Scheduling Yield	Clamp Step Clocks to Zero
$\bot$ (PASS)	POSIX 0	Authorized Execution	Zero-Copy Register Transfer

4.3 Tier 3: Bare-Metal Membrane (no_std Rust & eBPF)

The runtime verification membrane (`src/lifecycle.rs`) is compiled strictly under `#![no_std]` Rust and enforces zero dynamic heap allocation on the synchronous validation path:

$$M_{\text{alloc}} = 0 \quad \text{bytes} \tag{9}$$

Every inbound frame is clamped to fixed static stack allocations:

$$S_{\max} \leq 4096 \quad \text{bytes} \tag{10}$$

Execution is strictly bounded by deterministic worst-case execution time (WCET) ceilings:

$$\tau_{\text{override}} \leq 11.99 \quad \text{ms} \tag{11}$$

4.4 Tier 4: Fail-Closed Hardware Actuator Interlocks

At the physical boundary, actuation authority is isolated from software cognition:

Invariant 2 (**Non-Authorization State Conservation**)**.** Let $a \in \mathcal{A}$ be an action proposal evaluated by deterministic admissibility gate $D : \mathcal{A} \to \{\text{PASS}, \text{DENY}\}$. Then:

$$D(a) = \text{DENY} \implies \Delta_{\text{external}}^{\text{unauthorized}}(a) = 0 \tag{12}$$

Unauthorized execution guarantees zero external state mutation. An NMI interrupt drives physical relays to a safe-hold state within $\leq 100\,\mu$s.

5 Verification-Cost Inversion (VCI) & Automated CI

Traditional formal verification scales with high operational drag ($C_{\text{ops}} \gg 0$). DAE introduces **Verification-Cost Inversion (VCI)**, establishing an asymmetric economic barrier between automated machine verification and manual human refutation:

$$C_{\text{verify_machine}} \approx O(1) \quad [\text{25s CI Check}] \quad \lll \quad C_{\text{refute_human}} \approx O(\exp(N)) \tag{13}$$

Deterministic GitHub Actions Pipeline (`proof-ci.yml`)

All invariant theorems compile automatically in headless cloud environments under 25 seconds:

```
coqc proofs/dae_invariants.v
Print Assumptions fail_dominance_short_circuit.
# Verification Output: Closed under the global context (Assumptions = 0)
cargo test --manifest-path harness/Cargo.toml -- --nocapture
# Harness: 0 heap allocations (M_alloc = 0), latency <= 11.99ms
```

6 Domain Specialization & Empirical AI Diagnostics

The framework projects across three high-reliability domains in `autodidactic-credential-sovereignty`:

- **Aerospace Avionics**: TCM memory locks in Coq via `lia` replacing physical Triple Modular Redundancy (TMR) thermal overhead.
- **Zero-Knowledge Verification**: Coupling algebraic lattices to Circom constraint circuits (`dae_verifier.circom`).
- **Quantitative Finance**: Non-leakage transaction invariants enforcing atomic multi-party settlements.

Blind Cross-Engine Diagnostic Audit

Prompts stripped of branding were submitted to unassisted frontier reasoning models to evaluate structural recognition of the Regeneration Gap and the Single-Commit Invariant:

Table 3: Blind Model Diagnostic Audit Results

Reasoning Model	Diagnosed Failure Mechanism	Derived Invariant
Gemini	Nondeterministic Payload Drift	Dynamic Intent-UUID Isolation
Grok	Regenerative Payload Divergence	$\sum \text{CommittedEffects} \leq 1$
Copilot	Semantic Intent Re-materialization	$\sum \text{Effects(ExecID)} \leq 1$
ChatGPT	Generative Retry Divergence	$\sum_{e \in E(o)} \mathbf{1}[\text{Commit}(e)] \leq 1$

7 Archival Provenance & Canon Registry

SPECIFICATION COMPLETE: ARC-PUB-2026-DAE-01
DEPLOYMENT POSTURE: ARCHIVED PATH A CANONICAL EXTENSION
AUTHORIZATION HASH: `A-77-DELTA-SHIELD-LOCKED`

- **Core Reference Implementation**: https://github.com/trencinodin-stack/arcstone-continuity-core
- **Domain Specialization Engine**: https://github.com/trencinodin-stack/autodidactic-credential-sovereignty
- **Primary Zenodo DOI Anchor**: DOI 10.5281/zenodo.22665852
- **Academic Suite DOI**: DOI 10.5281/zenodo.23371581

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